

Qingyu Song

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WORK EXPERIENCE

Associate Professor, Xiamen University.	Mar. 2026
Assistant Professor, Xiamen University.	Sep. 2025

EDUCATION

The Chinese University of Hong Kong, Hong Kong, China Ph.D., Computer Science and Engineering Advisor: Prof. Hong Xu Thesis: Exploring the Foundation of Learning to Optimize: From Practice to Theory.	2021-2025
Tsinghua University, Beijing, China M.S., Control Engineering Advisor: Prof. Jianming Hu Thesis: Traffic Time Series Data Prediction with Graph Neural Networks.	2018-2021
Harbin Institute of Technology, Weihai, China B.S., Software Engineering Advisor: Dr. Xuefeng Piao Thesis: Vehicle Trajectory Cleaning and Traffic Flow Prediction with Deep Learning Methods.	2014-2018

RESEARCH INTEREST

My current research interests:

- Agent Framework. [P5]
- Zeroth-Order Optimization. [P3, C11]
- Learning to Optimize. [P1, C9, C8, C7, C6, C5, W2, W1]
- AI for Computer Network Management & Scheduling. [P8, C9, C8, C7, C5, W2, W1]
- Optimization for Deep Learning. [C11, C9, C8, P3, C11]
- Inference Acceleration for LLM. [P2]

My past research topics:

- Communication Efficient Federated Learning. [C4]
- Graph Neural Networks. [C5, W2, W1, C3, C1]
- Time Series Prediction. [C3, C1]

PUBLICATIONS

In submission

- P7. Mengrui Zhang#, Bang Huang#, Yunxin Xu, Haiying Huang, Luxi Zhao, Mochun Long, **Qingyu Song***, Qiao Xiang*, Xue Liu, Jiwu Shu, Toward Innovation Discovery in Networking Research.
- P6. Mingyuan Song, Huan Shen, Jinghui Jiang, Qiang Su, **Qingyu Song**, Lu Tang, Wanjian Feng, Fei Yuan, Qiao Xiang, Jiwu Shu. Argo: An efficient verification framework for distributed in-network computing. arXiv preprint arXiv:2511.08189
- P5. Yining Jiang, Yunxin Xu, Wenyun Xu, Yufan Zhu, Tangtang He, Haiying Huang, Letian Zhu, **Qingyu Song***, Qiang Su, Lizhao You, Lu Tang, Wanjin Feng, Yuchao Zhang, Linghe Kong, Qiao Xiang*, Jiwu Shu. Leveraging Large Language Models for Automated Reproduction of Networking Research Results. arXiv preprint arXiv:2509.21074 (2026)
- P4. Yao Wang, Kexin Yu, Wenyun Xu, Kaiqiang Hu, Ziyi Wang, Lizhao You, Qiang Su, Dong Guo, Haizhou Du, Wanjian Feng, **Qingyu Song**, Linghe Kong, Qiao Xiang, Jiwu Shu, Janus: Leveraging Incremental Computation for Efficient DNS Verification. arXiv preprint arXiv:2511.02559 (2025)
- P3. **Qingyu Song**, Peiyu Liao, Wenqian Zhao, Yiwen Wang, Shoubo Hu, Hui-Ling Zhen, Ning Jiang, and Mingxuan Yuan. Harnessing On-Device Large Language Model: Empirical Results and Implications for AI PC. arXiv preprint arXiv:2505.15030 (2025).
- P2. Wei Lin, **Qingyu Song**, and Hong Xu. The Multi-Query Paradox in Zeroth-Order Optimization. arXiv preprint arXiv:2509.15552 (2025)
- P1. Wei Lin, **Qingyu Song***, and Hong Xu. Adaptive Coordinate-Wise Step Sizes for Quasi-Newton Methods: A Learning-to-Optimize Approach. arXiv preprint arXiv:2412.00059 (2024).

Conferences, (* Corresponding Author)

- C12. Wei Lin, Yining Jiang, **Qingyu Song***, Qiao Xiang, Hong Xu, AGZO: Activation-Guided Zeroth-Order Optimization for LLM Fine-Tuning. In **ICML** 2026. (CCF A)
- C11. Siyong Huang, **Qingyu Song***, Lizhao You, Qiang Su, Lu Tang, Wanjian Feng, Fei Yuan, Qiao Xiang*, Jiwu Shu, REACT: Toward Real-Time, End-to-End, Adaptive Cross-layer Restoration for IP-Over-Optical Networks. In IEEE **IWQoS** 2026. (CCF B)
- C10. Mingjun Fang, Shuhao Zheng, Zonglun Li, Letian Zhu, **Qingyu Song***, Lizhao You, Qiang Su, Lu Tang, Wanjian Feng, Fei Yuan, Qiao Xiang*, Xue Liu, Jiwu Shu, TAegisPath: Privacy-Preserving Interdomain Data-Plane Verification with Versioned Verifiable Evidence. In ACM **APNet** 2026. (CCF C)
- C9. Qiang Su, Yining Jiang, Siyong Huang, **Qingyu Song***, Qiao Xiang*, Xue Liu, and Jiwu Shu. Toward Scalable and High-Performance GNN-Based Traffic Engineering with Free Path Selection. In IEEE **ICPADS** 2025. (CCF C)
- C8. **Qingyu Song**, Wei Lin, and Hong Xu. Learning Provably Improves the Convergence of Gradient Descent. In **NeurIPS** 2025. (CCF A)
- C7. Siyong Huang, **Qingyu Song**, Kexin Yu, Zhaoning Wang, Zhizhen Zhong, Qiao Xiang, and Jiwu Shu. Toward Scalable Learning-Based Optical Restoration. In ACM **APNet** 2025. (CCF C)
- C6. **Qingyu Song**, Wei Lin, Juncheng Wang, Hong Xu. Towards Robust Learning to Optimize with Theoretical Guarantees. In IEEE/CVF **CVPR**, 2024. (CCF A)
- C5. **Qingyu Song**, Juncheng Wang, Jingzong Li, Guocheng Liu, Hong Xu. A Learning-only Method for Multi-Cell Multi-User MIMO Sum Rate Maximization. In IEEE **INFOCOM**, 2024. (CCF A)

- C4. Yu Zhang, Wei Lin, Sisi Chen, **Qingyu Song**, Jiaxun Lu, Yunfeng Shao, Bei Yu, Hong Xu. Fed2Com: Towards Efficient Compression in Federated Learning. In IEEE **ICNC**, 2024.
- C3. **Qingyu Song**, RuiBo Ming, Jianming Hu, Haoyi Niu, Mingyang Gao. Graph Attention Convolutional Network: Spatiotemporal Modeling for Urban Traffic Prediction. In IEEE **ITSC**, 2020. (Tsinghua A)
- C2. Jinhua Chen, **Qingyu Song**, Can Zhao, Zhiheng Li. Graph Database and Relational Database Performance Comparison on a Transportation Network. In **ICACDS**, 2020.
- C1. **Qingyu Song**, Jianming Hu, Ruobing Zhang, Zuo Zhang. An Urban Topological Map Generation Method for Traffic Flow Prediction Based on Road Segment Clustering with Floating Vehicle Trajectory Dataset. In COTA **CICTP**, 2019. (Tsinghua B)

Workshops

- W2. **Qingyu Song**, Guocheng Liu, Hong Xu. Learning to Optimize Non-Convex Sum-Rate Maximization Problems. In **ICML** 2023, 1st Workshop on Synergy of Scientific and Machine Learning Modeling.
- W1. **Qingyu Song**, Guocheng Liu, Hong Xu. Towards a Learning-Only Approach for Non-Convex Sum Rate Maximization. In ACM **SigMetrics** 2023, 1st Workshop on Learning-augmented Algorithms: Theory and Applications.

RESEARCH EXPERIENCE

- Jan. 2025 - Apr. 2025 Intern, Huawei Noah's Ark Lab. Mentor: Dr. Hui-Ling Zhen, Dr. Shoubu Hu
Project: LLM Quantization.
- Mar. 2024 - Jan. 2025 CUHK Advisor: Prof. Hong Xu
Project: Learning Provably Improves the Convergence of Gradient Descent.
- Sep. 2023 - Dec. 2023 CUHK Advisor: Prof. Hong Xu, Prof. Juncheng Wang (HKBU)
Project: Convergence Analysis of Learning to Optimize (L2O) in Out-of-Distribution (OOD) Scenarios
- Nov. 2022 - May. 2023 Visiting Researcher, Huawei Noah's Ark Lab Mentor: Dr. Guochen Liu
Project: Learning-Based Precoding for Multi-Cell, Multi-User MIMO Interference Reduction.
- Nov. 2020 - May. 2021 Research Assistant, Tsinghua University Advisor: Prof. Jianming Hu
Project: National Key R&D Program, 5.1 Efficient and Intelligent Vehicle-to-Vehicle Networking Technology for Tokyo Olympics, Topic 2 - Research on Traffic State Perception System.
- Jan. 2020 - May. 2020 Research Assistant, Tsinghua University Advisor: Prof. Jianming Hu
Project: Graph Neural Network-based Traffic Flow Prediction.
- Mar. 2018 - May. 2018 UG Research Assistant, Tsinghua University Advisor: Prof. Jianming Hu
Project: Vehicle Traffic Trajectory Data Cleaning and Augmentation.
- Oct. 2017 - Jan. 2018 R&D Intern, NEBULA-LINK Internet Technology Co., Ltd. Mentor: Dr. Yizhi Wang
Project: Android App Development and Data Analysis for Advanced Driver Assistance Systems.
- May. 2016 - May. 2017 Part-time R&D Intern, HITWH Mentor: Dr. Xuefeng Piao
Project: Android App Development for Inspection System with Client/Server Architecture.

TEACHING

Fall 2025	XMU CSCI U1000052, Introduction to Computational Thinking
Spring 2022	CUHK CSCI 4430 / ESTR 4120, Data Communication and Computer Networks
Spring 2021	CUHK CSCI 4430, Data Communication and Computer Networks
Fall 2021	CUHK ENGG 2760A / ESTR 2018: Probability for Engineers

AWARDS

Dec. 2025 IEEE ICPADS Best Paper
Oct. 2025 NeurIPS 2025 Top Reviewer
Sep. 2025 2025 ACM SIGCSE China Doctoral Dissertation Award
Jun. 2025 Travel Grant, ACM APNet 2025
Mar. 2024 Student Travel Grant, IEEE INFOCOM 2024
Jul. 2023 Financial Support, ICML 2023
2021 - 2025 Full Postgraduate Studentship, CUHK
2019 - 2020 First Honor and Second Honor Scholarships, Tsinghua University
Jun. 2018 Outstanding Graduate Award at Provincial Level, People's Government of Shandong Province
2015 - 2017 First Honor and Second Honor Scholarships, Harbin Institute of Technology, Weihai

SERVICES

Conference Reviewer: ICML 2026, ICLR 2026, NeurIPS 2025, ICML 2025, ICLR 2025.
PC Member: IJCAI-ECAI 2026, AAAI 2026, EuroSys 2026 (Shadow), IJCAI 2025, ECAI 2025, ICA3PP 2025, ICPADS 2025.
Journal Reviewer: TNSE.

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